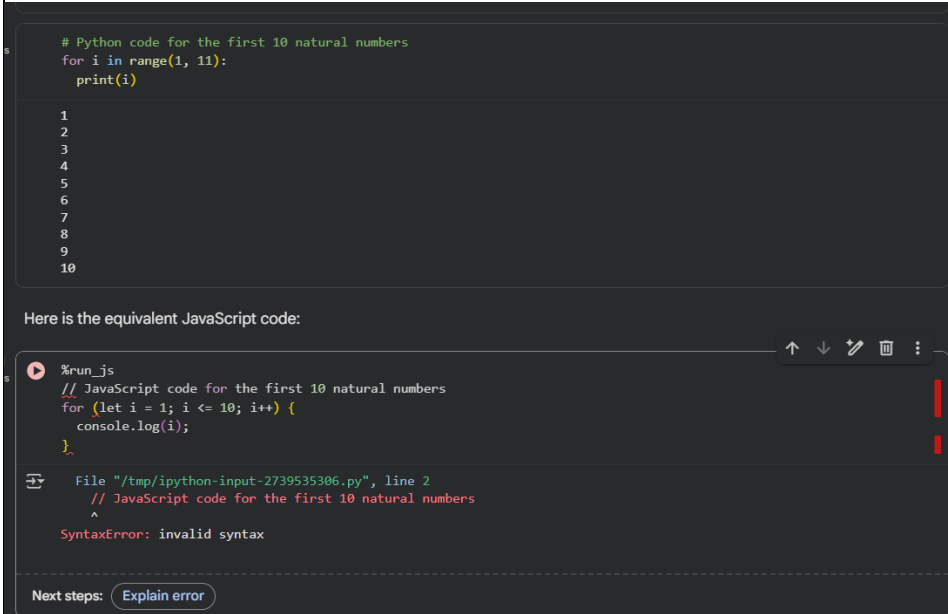
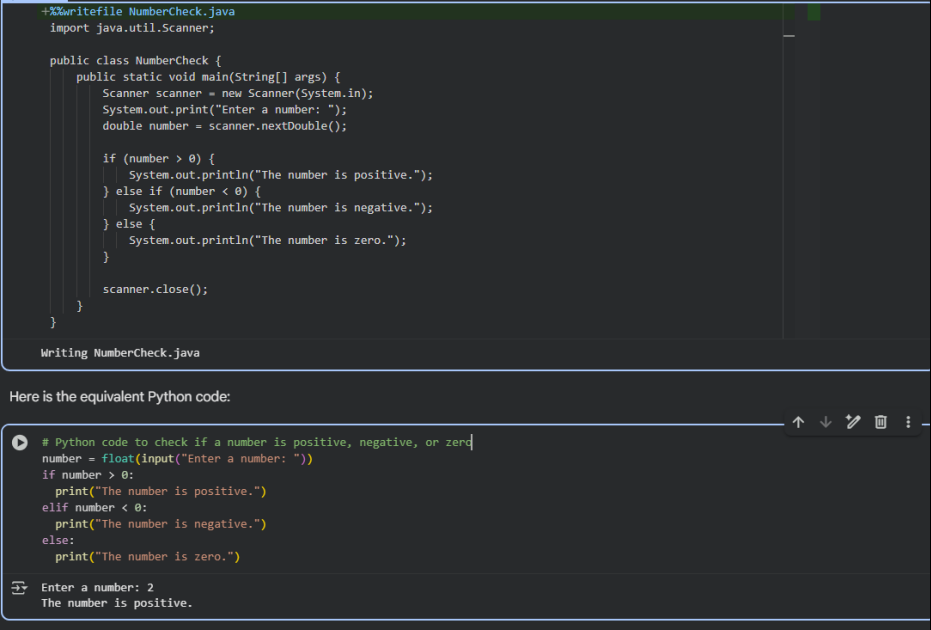


SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
Course Coordinator Name		Venkataramana Veeramsetty	
Instructor(s) Name		Dr. V. Venkataramana (Co-ordinator)	
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		Dr. Brij Kishor Tiwari	
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		Dr. Mohammand Ali Shaik	
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		Mr. B.Raju	
		Intern 1 (Dharma teja)	
		Intern 2 (Sai Prasad)	
		Intern 3 (Sowmya)	
		NS_2 (Mounika)	
Course Code	24CS002PC215	Course Title	AI Assisted Coding
Year/Sem	II/I	Regulation	R24
Date and Day of Assignment	Week10 - Thursday	Time(s)	
Duration	2 Hours	Applicable to Batches	
AssignmentNumber:19.4(Present assignment number)/24(Total number of assignments)			
Q.No.	Question		Expected Time to complete
1	Lab 19 – Code Translation: Converting Between Programming Languages Lab Objectives: Understand how AI tools can assist in translating code between different programming languages.		Week10 - Thursday

	• Learn to verify correctness and functionality after translation. • Explore syntactic and semantic differences between languages (e.g., Python, Java, C++). • Practice debugging and optimizing AI-translated code.	
	Task 1: Translate a Simple Program (Python → JavaScript) • Instructions: • Write a Python function <code>print_numbers()</code> that prints the first 10 natural numbers using a loop. • Translate the function into JavaScript as a reusable function <code>printNumbers()</code>. • Call the function in both languages to display results. • Expected Output: • 1 • 2 • 3 • ...10 Prompt : Translate the following Python code of 1st 10 natural numbers into equivalent JavaScript code. The JavaScript version should have the same logic, output, and structure.(Write a python code for 1st 10 natural numbers and then proceed with translation)  The screenshot shows a code editor with the following content: <pre># Python code for the first 10 natural numbers for i in range(1, 11): print(i)</pre> Below the Python code, the first 10 natural numbers are listed vertically: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Below the list, the text "Here is the equivalent JavaScript code:" is displayed. Below the text, the JavaScript code is shown: <pre>%run_js // JavaScript code for the first 10 natural numbers for (let i = 1; i <= 10; i++) { console.log(i); }</pre> Below the JavaScript code, a syntax error message is displayed: <pre>File "/tmp/ipython-input-2739535306.py", line 2 // JavaScript code for the first 10 natural numbers ^ SyntaxError: invalid syntax</pre> At the bottom of the screenshot, there is a "Next steps:" section with a button labeled "Explain error".	
	Task 2: Convert Conditional Statements (Java → Python) • Instructions:	

	• Write a Java method checkNumber(int num) that checks if a number is positive, negative, or zero. • Translate the method into a Python function check_number(num). • Call the function/method with different inputs and compare outputs. • Expected Output: ○ Input: -5 → Output: The number is negative ○ Input: 0 → Output: The number is zero ○ Input: 7 → Output: The number is positive Prompt : Write a java code to check whether user input number is positive, negative or zero. Convert java code into python  The screenshot shows two code snippets. The top snippet is a Java program named 'NumberCheck.java' that uses a Scanner to read a double value and prints 'The number is positive.', 'The number is negative.', or 'The number is zero.' based on its value. The bottom snippet is the equivalent Python code, which uses float(input()) to read a number and prints the same messages. Below the Python code, a terminal window shows the input '2' and the output 'The number is positive.'	
	Task 3: Translate Recursive Function (Python → C++) • Instructions: • Write a Python function factorial(n) that calculates factorial of a number using recursion. • Translate the same into a C++ function int factorial(int n). • Call the function in both languages with inputs 5 and 0. • Expected Output: ○ Input: 5 → Output: Factorial = 120 ○ Input: 0 → Output: Factorial = 1 Prompt : Write a recursive function in python to calculate factorial of a	

number .Convert the code into c++code

```
# Recursive function to calculate factorial in Python
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n-1)

# Example usage
num = 5
print(f"The factorial of {num} is {factorial(num)}")

The factorial of 5 is 120
```

Here is the equivalent C++ code:

```
%writefile main.cpp
#include <iostream>

// Recursive function to calculate factorial in C++
long long factorial(int n) {
    if (n == 0)
        return 1;
    else
        return n * factorial(n - 1);
}

int main() {
    int num = 5;
    std::cout << "The factorial of " << num << " is " << factorial(num) << std::endl;
    return 0;
}
```

Writing main.cpp

Task 4: Data Structures with Functions (JavaScript → Python)

- **Instructions:**

- Write a JavaScript function printStudents(students) that takes an array of student names and prints each name.
- Translate it into a Python function print_students(students) using a list.
- Test both functions with sample student names.

- **Expected Output:**

- Student List:
- Alice
- Bob
- Charlie

Prompt :

Write a javascript program to list student names and print them , convert the java script function into python using lists

```
%run_js
// JavaScript program to list and print student names
let studentNames = ["Alice", "Bob", "Charlie", "David"];

console.log("Student Names:");
for (let i = 0; i < studentNames.length; i++) {
  console.log(studentNames[i]);
}
```

```
File "/tmp/ipython-input-995876762.py", line 2
// JavaScript program to list and print student names
^
SyntaxError: invalid syntax
```

Next steps: [Explain error](#)

Here is the equivalent Python code:

```
# Python program to list and print student names using lists
student_names = ["Alice", "Bob", "Charlie", "David"]

print("Student Names:")
for name in student_names:
  print(name)
```

Student Names:
Alice
Bob
Charlie
David

Task 5: Class & Object Translation (Python → Java)

- **Instructions:**

1. Write a **Python class** Car with attributes: brand, model, year.
2. Add a **method** display_details() that prints car details.
3. Translate the same into a **Java class** Car with attributes and a **method** displayDetails().
4. Create an object in both languages and call the method.

- **Expected Output:**

- Car Details:
- Brand: Toyota
- Model: Corolla
- Year: 2020

Prompt : Write a python class to represent car with attributes and method to display details. Now , trnslate the python class into java

```
# Python class to represent a car
class Car:
    def __init__(self, make, model, year):
        self.make = make
        self.model = model
        self.year = year

    def display_details(self):
        print(f"Make: {self.make}, Model: {self.model}, Year: {self.year}")

# Example usage
my_car = Car("Toyota", "Corolla", 2020)
my_car.display_details()
```

Make: Toyota, Model: Corolla, Year: 2020

Here is the equivalent Java code:

↳ Gemini

```
%Writefile Car.java
// Java class to represent a car
public class Car {
    String make;
    String model;
    int year;

    // Constructor
    public Car(String make, String model, int year) {
        this.make = make;
        this.model = model;
        this.year = year;
    }

    // Method to display details
    public void displayDetails() {
        System.out.println("Make: " + this.make + ", Model: " + this.model + ", Year: " + this.year);
    }

    // Main method for example usage (optional, but good for testing)
    public static void main(String[] args) {
        Car myCar = new Car("Honda", "Civic", 2023);
        myCar.displayDetails();
    }
}
```

Writing Car.java

✓ Deliverables (For All Tasks)

1. AI-generated prompts for code and test case generation.
2. At least 3 assert test cases for each task.
3. AI-generated initial code and execution screenshots.
4. Analysis of whether code passes all tests.
5. Improved final version with inline comments and explanation.
6. Compiled report (Word/PDF) with prompts, test cases, assertions, code, and output.